FYS2160 – Thermodynamics and Statistical Physics

Oblig 2

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1-THE EINSTEIN CRYSTAL

A:

General formula for the multiplicity

$$\Omega(N,q) = \binom{q+N-1}{q} = \frac{(q+N-1)!}{q!(N-1)!} \tag{1}$$

We have two Einstein crystals : A with N_A oscillators and q_A units of energy. B with N_B oscillators and q_B units of energy.

The multiplicity of A is:

$$\Omega_A(N_A,q_A) = \binom{q_A+N_A-1}{q_A} = \frac{(q_A+N_A-1)!}{q_A!(N_A-1)!} \tag{2}$$

The multiplicity of B is:

$$\Omega_B(N_B,q_B) = \binom{q_B+N_B-1}{q_B} = \frac{(q_B+N_B-1)!}{q_B!(N_B-1)!} \tag{3}$$

The total multiplicity when A and B in thermal contact is the product of their multiplicities:

$$\Omega_{total} = \Omega_A(N_A,q_A) \times \Omega_B(N_B,q_B) \tag{4}$$

The probability is equal to the individual multiplicity divided by the sum of all multiplicities at the thermal equilibrium:

$$P(q_A) = \frac{\Omega(N_A,q_A)}{\sum \Omega_A \Omega_B} \tag{5}$$

and

$$P(q_B) = \frac{\Omega(N_B, q_B)}{\sum \Omega_A \Omega_B}$$

The entropy can be found by the formula

$$S = k \ln(\Omega)$$

The temperature T is the partial derivative of the entropy S as function of energy U at constant volume.

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V,N}$$

or :

$$T = \left(\frac{\partial S}{\partial U} \right)^{-1}_{V,N}$$

B:

a:

$N = 3$, $q = 3$. The possible microstate of the system is :

Microstate	
1	{0, 0, 3}
2	{0, 3, 0}
3	{3, 0, 0}
4	{1, 1, 1}
5	{1, 0, 2}
6	{1, 2, 0}
7	{0, 2, 1}
8	{0, 1, 2}
9	{2, 0, 1}
10	{2, 1, 0}

We can see that the total microstates is 10.

b:

From equation **(1)**:$q = 3$, $N = 3$:

$$\Omega(3,3) = \binom{3+3-1}{3} = \binom{5}{3} = \frac{(5)!}{3!2!} = 10$$

. which agrees with what we gotin **a:**

c:

For a system consisting of subsystem A with $N_A = 2$ and $q_A = 5$ and subsystem B with $N_B = 2$ and $q_B = 1$.

$$N_{tot} = 4$$
$$q_{tot} = 6$$

Allpossible microstates :

q_A 1	q_A 2	q_B 1	q_B 2
5	0	1	0
5	0	0	1
0	5	0	1
0	5	1	0
1	4	0	1
1	4	1	0
4	1	0	1
4	1	1	0
2	2	0	1
2	2	1	0
3	2	0	1
3	2	1	0
2	3	1	0
2	3	0	1

d:

Each of q_A,q_B can take all the values from $(0, 6)$

as pairs at the same time :

$$(q_A, q_B = q - q_A)$$
$$(0, 6)$$
$$(1, 5)$$
$$(2, 4)$$
$$(3, 3)$$
$$(4, 2)$$
$$(5, 1)$$

$$(6, 0)$$

The number of combinations is: $N + 1$

e:

Weuse equations **(2)**,**(3)** to calculate the mulytiplicity of each crystal and **(4)** to calculate the total Ω

In [1]:

```python
from math import comb

def microstates(q, N):
    return comb(q + N - 1, q)

def compatible_microstates(q_total, N_A, N_B):
    macros = []
    q_A_values = []
    q_B_values = []
    for q_A in range(q_total + 1):
        q_B = q_total - q_A
        q_A_values.append(q_A)
        q_B_values.append(q_B)
        omega_tot = microstates(q_A, N_A) * microstates(q_B, N_B)
        macros.append(omega_tot)
    return q_A_values, q_B_values, macros

q_A_values, q_B_values, macro_states = compatible_microstates(q_total=6, N

print("The pairs (q_A,q_B):")
both = zip(q_A_values, q_B_values)
for A, B in both:
    print(f"(q_A,q_B):({A}, {B})")
print("\n")
print(f"The list of  macrostates: {macro_states}")

total_microstates = sum(macro_states)
print(f"The total microstates: {total_microstates}")
print("\n")

both = list(zip(q_A_values, q_B_values))
the_combimnations = zip(both,macro_states)
for m,M in the_combimnations:
    print(f"For the pair :{m}, the correspond macrostate is :{M}")
```

```
The pairs (q_A,q_B):
(q_A,q_B):(0, 6)
(q_A,q_B):(1, 5)
(q_A,q_B):(2, 4)
(q_A,q_B):(3, 3)
(q_A,q_B):(4, 2)
(q_A,q_B):(5, 1)
(q_A,q_B):(6, 0)


The list of  macrostates: [7, 12, 15, 16, 15, 12, 7]
The total microstates: 84


For the pair :(0, 6), the correspond macrostate is :7
For the pair :(1, 5), the correspond macrostate is :12
For the pair :(2, 4), the correspond macrostate is :15
For the pair :(3, 3), the correspond macrostate is :16
For the pair :(4, 2), the correspond macrostate is :15
For the pair :(5, 1), the correspond macrostate is :12
For the pair :(6, 0), the correspond macrostate is :7
```

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For the probabilities, we use equation **(5)**:

In [2]:
```
probabilities = []

for i in macro_states:
    p = i / total_microstates
    probabilities.append(p)
print("\n")
Macrostates_probabilities = zip(both,macro_states,probabilities)
for m,M,P in Macrostates_probabilities:
    print(f"The pair:{m}, the macrostate is:({M}) with probability = {P:.4
```

```
The pair:(0, 6), the macrostate is:(7) with probability = 0.0833
The pair:(1, 5), the macrostate is:(12) with probability = 0.1429
The pair:(2, 4), the macrostate is:(15) with probability = 0.1786
The pair:(3, 3), the macrostate is:(16) with probability = 0.1905
The pair:(4, 2), the macrostate is:(15) with probability = 0.1786
The pair:(5, 1), the macrostate is:(12) with probability = 0.1429
The pair:(6, 0), the macrostate is:(7) with probability = 0.0833
```

f:

Before the two crystals became in thermal contact, we calculate the total microstates Ω_{tot} by applying equations **(2)** ,**(3)**, **(4)**:

$$\Omega_A(N_A, q_A) = \binom{6}{5} = 6$$
$$\Omega_B(N_B, q_B) = \binom{2}{1} = 2$$
$$\Omega_{tot} = \Omega_A \times \Omega_B = 12$$

After they became in thermal contact, the total number of microstates is the sum of the products of the microstates for each macrostate of each pair (q_A, q_B) :

$$\Omega_{tot} = \sum \Omega_A \Omega_B = 84$$

the total number ov microstates after cntact is greater than before contact.

(g):

In [5]:
```
import matplotlib.pyplot as plt
import numpy as np

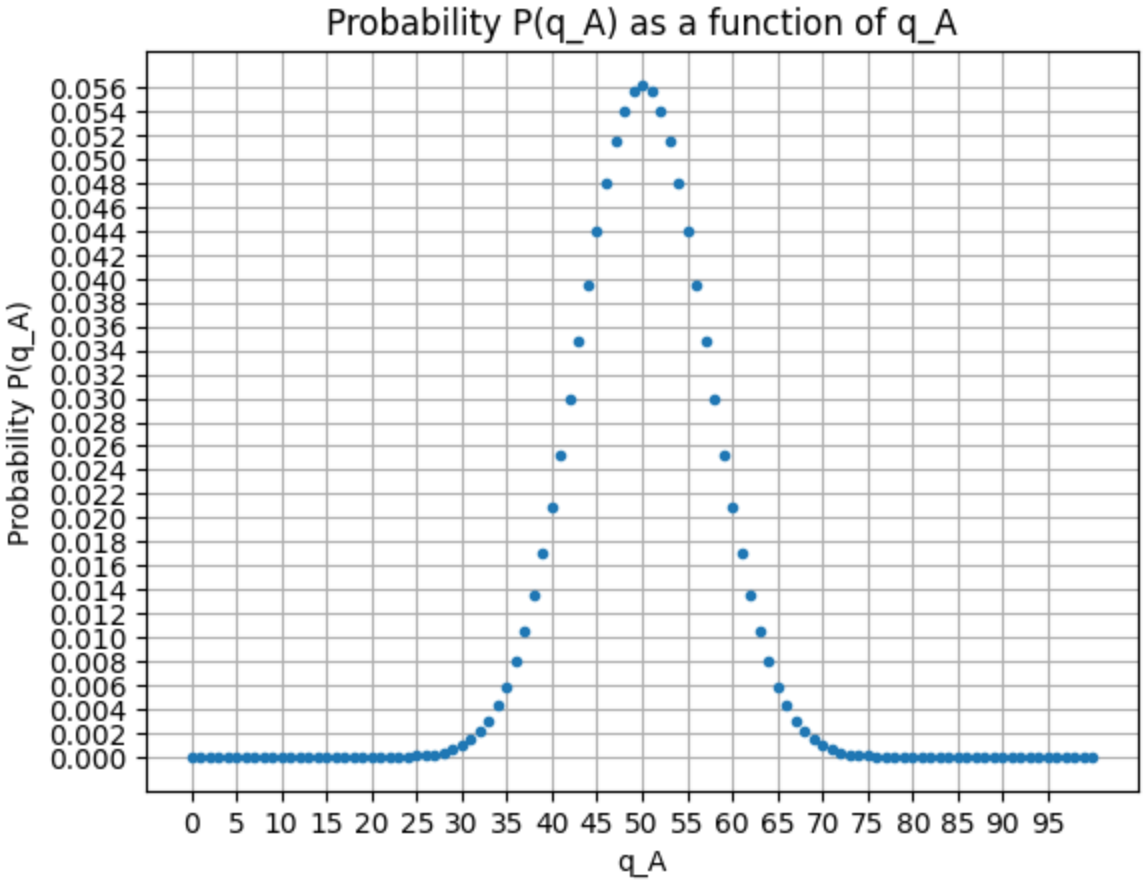

q_A_values, q_B_values, macro_states = compatible_microstates(q_total=100,
total_microstates = sum(macro_states)

probabilities = []
for i in macro_states:
    p = i / total_microstates
    probabilities.append(p)

```

In [49]:
```
plt.plot(q_A_values, probabilities, ".")
plt.xticks(np.arange(min(q_A_values), max(q_A_values) , 5))
plt.yticks(np.arange(min(probabilities), max(probabilities) , 0.002))

plt.xlabel("q_A")
plt.ylabel("Probability P(q_A)")
plt.title("Probability P(q_A) as a function of q_A")
plt.grid()
plt.show()
```



In [26]:
```
P_high = max(probabilities)
q_A_index = probabilities.index(P_high)
print(f"The most probable q_A is : ({q_A_index}) with probability P({q_A_i
```

```
The most probable q_A is : (50) with probability P(50)= 0.0562.
```

We see from the plot that the most probable macrostate correspond to the pair $(q_A, q_B) = (50, 50)$ with probability $P = 0.0562$.

C:

(a)

$$\Omega(N, q) = \binom{q + N - 1}{q} = \frac{(q + N - 1)!}{q!(N - 1)!}$$

$$\ln \Omega(N, q) = \ln\left(\frac{(q + N - 1)!}{q!(N - 1)!}\right)$$

We have that $N \gg 1$, we can simplify the term $N - 1 \approx N$

$\ln \Omega(N, q)$in this case will become :

$$\ln \Omega(N, q) \approx \ln\left(\frac{(q + N)!}{q!N!}\right)$$

$$\ln \Omega(N, q) \approx \ln(q + N)! - \ln q! - \ln N!$$

From Stirling's approximation ,we can write $\ln(N!) = N \ln N - N$, we get :

$$\ln \Omega(N, q) \approx (q + N) \ln(q + N) - q - N - q \ln q + q - N \ln N + N$$

The terms $q, -q$, $N, -N$cancel out :

$$\ln \Omega(N, q) \approx (q + N) \ln(q + N) - q \ln q - N \ln N \tag{6}$$

$$\ln(q + N) = \ln\left(q(1 + \frac{N}{q})\right) = \ln q + \ln(1 + \frac{N}{q})$$

$\ln(1 + \frac{N}{q}) \approx \frac{N}{q}$ when $\frac{N}{q} \ll 1$

$\Longrightarrow$,

$$\ln(q + N) \approx \ln q + \frac{N}{q}$$

We substitute in **(6)**:

$$\ln \Omega(N, q) \approx (q + N)(\ln q + \frac{N}{q}) - q \ln q - N \ln N$$

$$\ln \Omega(N, q) \approx (q + N) \ln q + (q + N)\frac{N}{q}) - q \ln q - N \ln N$$

$$\ln \Omega(N, q) \approx q \ln q + N \ln q + N + \frac{N^2}{q} - q \ln q - N \ln N$$

We have also , $\frac{N}{q} \ll 1$, the term $\frac{N^2}{q}$ can be negligible. canceling out the terms $q \ln q$ and $-q \ln q$, we get :

$$\ln \Omega(N, q) \approx N \ln q + N - N \ln N \approx N \ln \frac{q}{N} + N \approx N(\ln \frac{q}{N} + 1)$$

(b)

The entropy is given by :

$$S = k \ln(\Omega)$$

$$S = k \ln(\Omega) = Nk(\ln \frac{q}{N} + 1)$$

(c)

The temperature T is the partial derivative of the entropy S as function of energy U at constant volume.

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{V,N}$$

We can rewrite the expression for entropy as function of ebnergy, $q = \frac{U}{\epsilon}$

$$S = NK \ln \frac{U}{N\epsilon} + NK$$

$$\frac{1}{T} = \left(\frac{\partial (NK \ln \frac{U}{N\epsilon} + NK)}{\partial U} \right)_{V,N} = NK \frac{N^2\epsilon^2}{UN^2\epsilon^2} = \frac{NK}{U}$$

$\implies$,

$$U = NKT$$

Equipartition theorem says that $U = \frac{1}{2} f N K T$, Einstein crystal has two degrees of reedom $f = 2$, the resultagrees with Equipartition theorem.

1-TTHE SPIN SYSTEM

(a)

In a pin system , we have two possible states : spin up or spin down . For N spin systems ,the total microstates will be 2^N

(b)

S_+ is the number of spin up, $S_-$is the number of spin down . The energy $U_+ = -\mu B$ for the spin up and $U_- = \mu B$ for spin down.

fpr S_+ spin down , the energy will be. $U_+ = -S_+ \mu B$ and for S_- spin down, the energy will be $U_+ - = S_- \mu B$

The total energy is the sum of both energies :

$$U = U_- + U_+ = S_- \mu B - S_+ \mu B = \mu B(S_- - S_+) = -2\mu B s$$

(c)

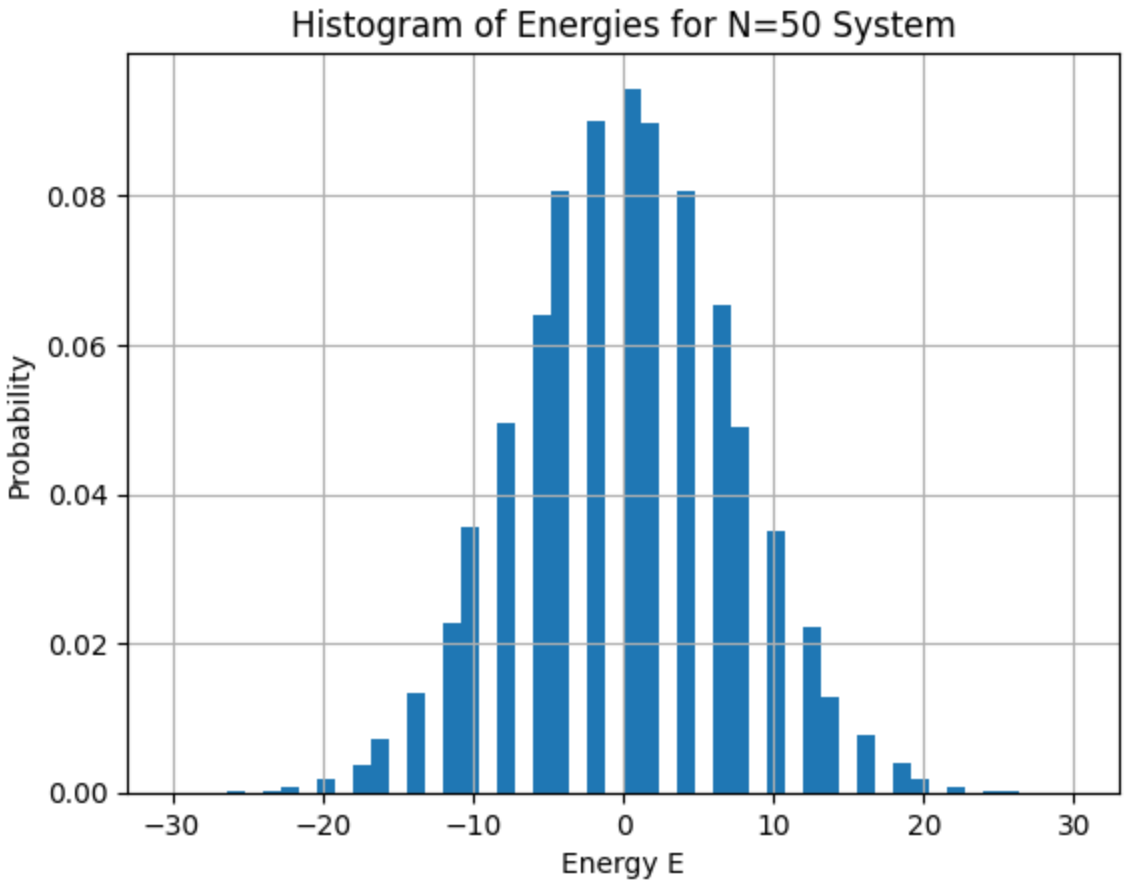
In [33]:

```python
import numpy as np
import matplotlib.pyplot as plt


M = 100000
N = 50
s_values = []
energies = []

muB = 1
for _ in range(M):
    spins = np.random.choice([-1, 1], size=N)
    s = sum(spins) / 2
    s_values.append(s)
    E = -2 * muB * s
    energies.append(E)


plt.hist(energies, bins=50, density=True)
plt.xlabel("Energy E")
plt.ylabel("Probability")
plt.title("Histogram of Energies for N=50 System")
plt.grid()
plt.show()
```



(d)

$$\Omega(N, S_+) = \binom{N}{S_+} = \frac{N!}{S_+!(N - S_+)!}$$

We have that $S_+ + S_- = N \implies N - S_+ = S_-$

$$\Omega(N, S_+) = \binom{N}{S_+} = \frac{N!}{S_+!S_-!}$$

(e)

From the previous equation :

$$\Omega(N, S_+) = \binom{N}{S_+} = \frac{N!}{S_+!S_-!}$$

$2s = S_+ - S_-$ and $N = S_+ + S_- \implies 2s + N = 2S_+, \implies$

$$S_+ = s + \frac{N}{2}$$

$$N_- = N - N_+ = \frac{N}{2} - s$$

We insert these two expression into $\Omega(N, S_+)$, we get:

$$\Omega(N, s) = \binom{N}{s} = \frac{N!}{(s + \frac{N}{2})!(\frac{N}{2} - s)!}$$

(F)

The only similar problem i founf is the multiplicity of Large Ensetein solids.

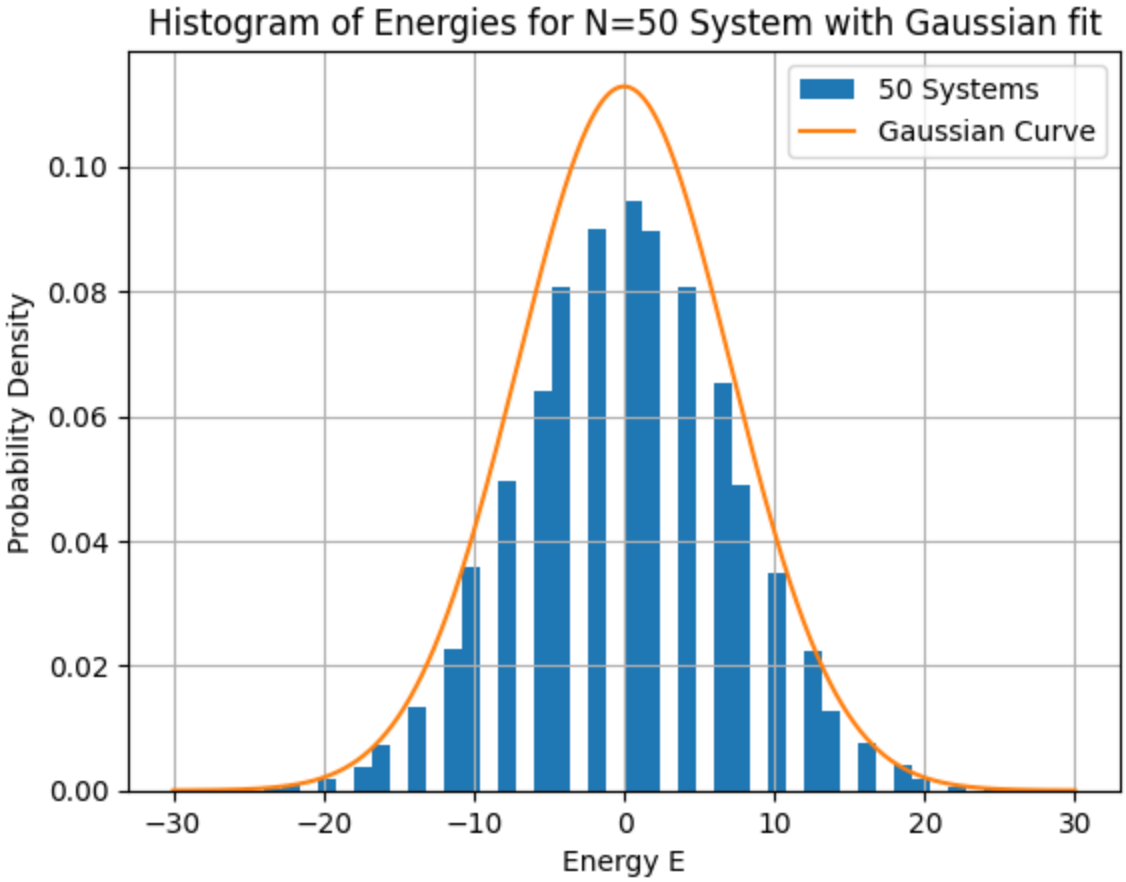
The constant $\Omega(N, 0)$ in our problem here,is the probability of having zero net spin out of N spins, and the exponential term $e^{\frac{-2s^2}{N}}$ is equivilant to the $\Omega(N, s)$

(g)

In [41]:
```
plt.hist(energies, bins=50, density=True,label="50 Systems")
plt.xlabel("Energy E")


x = np.linspace(min(energies), max(energies), 10000)
gaussian_curve = (1/(np.sqrt(2 * np.pi * (N/4)))) * np.exp(-2 * (x/(2*muB)
plt.plot(x, gaussian_curve, label="Gaussian Curve")

plt.xlabel("Energy E")
plt.ylabel("Probability Density")
plt.title("Histogram of Energies for N=50 System with Gaussian fit")
plt.legend()
plt.grid()
plt.show()
```



(h)

$$S = k\ln(\Omega) = Nk(\ln\frac{q}{N} + 1)$$

$$\Omega(N, S_+) = \binom{N}{S_+} = \frac{N!}{S_+!(N - S_+)!}$$

$$S = k\ln(\Omega) = k\ln\left(\frac{N!}{S_+!(N - S_+)!}\right)$$

$$S = k\left(\ln N! - \ln S_+! - \ln(N - S_+)!\right)$$

$$S = k\left(N \ln N - N - S_+ \ln S_+ + S_+ - (N - S_+ \ln(N - S_+) + N - S_+)\right)$$

$$S = k\left(N \ln N - S_+ \ln S_+ - (N - S_+)\ln(N - S_+)\right))$$

(i)

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} = \frac{\partial S}{\partial S_+}\frac{\partial S_+}{\partial U}\right)_{V,N}$$

We write S_+ as function of U:

$$U = -S_+\mu B \implies S_+ = \tfrac{U}{\mu B} \implies \tfrac{\partial S_+}{\partial U} = -\tfrac{1}{\mu B}$$

$$\frac{1}{T} = \left(-\frac{1}{\mu B}\frac{\partial S}{\partial S_+}\right)_{V,N}$$

$$\frac{\partial S}{\partial S_+} = k(-\ln s_+ - 1 + \ln(N - S_+) + 1)$$

$$\frac{\partial S}{\partial S_+} = k(-\ln s_+ + \ln(N - S_+)) = k\ln(\frac{N - S_+}{S_+})$$

$$\frac{1}{T} = -\frac{1}{\mu B}k\ln(\frac{N - S_+}{S_+})$$

$$T = -\frac{\mu B}{k\ln(\frac{N-S_+}{S_+})}$$

In [ ]:
```
1
```